

UTENA, Lithuania

WMO: 266330

Lat: 55.5153N

Lon: 25.5897E

Elev: 106

StdP: 100.06

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-16.8	-23.6	0.5	-20.3	-19.8	0.7	-15.9	7.2	0.3	6.4	-0.5	0.8	140	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.6	20.0	26.7	18.9	25.0	18.1	21.2	26.6	20.1	25.0	19.0	23.7	2.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	24.2	18.2	13.2	22.6	17.2	12.4	21.6	61.8	26.7	57.9	25.1	54.5	23.7	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.4	4.8	-24.3	31.4	4.8	1.7	-27.7	32.6	-30.5	33.6	-33.2	34.5	-36.7	35.7		
(5)			-24.5	22.8	4.7	0.9	-27.9	23.5	-30.6	24.0	-33.2	24.5	-36.6	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-4.2	-3.9	0.2	7.0	12.0	15.6	18.0	16.9	12.3	6.8	2.2	-1.7
	DBStd	9.01	6.19	6.21	4.60	4.35	4.29	3.43	3.08	3.09	3.43	3.92	4.48	5.37
	HDD10.0	1998	441	389	303	111	27	1	0	0	17	113	233	362
	HDD18.3	4274	699	622	561	341	202	96	44	65	183	357	483	620
	CDD10.0	842	0	0	1	21	89	169	248	215	86	14	0	0
	CDD18.3	76	0	0	0	0	5	14	34	22	1	0	0	0
	CDH23.3	779	0	0	0	5	56	154	331	221	14	0	0	0
	CDH26.7	165	0	0	0	0	6	26	84	48	1	0	0	0
Wind	WSAvg	2.2	2.5	2.4	2.5	2.3	2.1	2.0	1.8	1.7	1.9	2.3	2.5	2.6
Precipitation	PrecAvg	705	49	44	40	39	61	74	79	81	61	67	55	52
	PrecMax	876	111	74	76	74	163	144	196	141	146	132	103	80
	PrecMin	538	11	17	13	6	7	5	7	4	16	13	2	24
	PrecStd	93	20	15	17	20	29	32	46	39	33	30	23	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	6.7	14.6	24.0	27.5	29.3	31.7	30.7	25.3	18.3	11.9	8.8
		MCWB	6.2	4.7	8.3	15.1	18.4	19.6	21.8	21.3	17.4	13.9	9.8	7.6
	2%	DB	5.2	5.1	11.0	20.8	24.9	27.1	29.0	27.7	22.4	15.8	10.3	7.2
		MCWB	4.2	3.7	6.5	12.9	16.8	19.0	20.6	19.6	16.5	12.8	8.9	6.0
	5%	DB	3.7	4.0	8.3	18.1	22.7	25.1	26.9	25.8	20.1	14.0	8.9	5.7
		MCWB	2.8	2.9	5.3	11.1	15.4	18.0	19.6	18.4	15.3	11.8	7.7	4.8
	10%	DB	2.5	3.0	6.3	15.4	20.5	23.1	24.9	23.7	18.3	12.6	7.8	4.3
		MCWB	1.8	2.0	4.1	9.7	14.2	16.8	18.7	17.7	14.6	10.7	6.8	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	5.3	9.0	15.8	19.6	21.6	23.1	22.4	18.9	14.7	10.3	7.8
		MCDB	7.3	6.3	13.3	22.5	25.7	27.1	29.4	28.2	23.2	17.4	11.2	8.6
	2%	WB	4.3	3.9	7.1	13.4	17.9	19.9	21.6	20.7	17.3	13.2	9.0	6.1
		MCDB	5.1	4.8	9.8	19.7	23.5	25.1	27.1	26.1	21.0	15.2	10.1	6.9
	5%	WB	2.9	3.0	5.7	11.7	16.3	18.7	20.4	19.5	16.0	12.0	7.8	4.8
		MCDB	3.6	3.9	7.9	16.6	21.4	23.8	25.3	24.1	19.1	13.7	8.8	5.6
	10%	WB	1.8	2.1	4.3	10.3	14.8	17.4	19.3	18.3	14.9	10.9	6.8	3.4
		MCDB	2.5	3.0	6.2	14.6	19.4	22.0	23.8	22.5	17.7	12.5	7.7	4.2
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.5	7.2	10.3	11.2	10.3	10.2	10.6	9.1	6.3	3.8	3.6
		MCDBR	3.8	4.2	10.5	15.3	14.1	13.7	13.7	14.0	12.4	8.2	4.8	3.8
		MCWBR	3.4	3.5	6.9	8.4	7.0	6.8	6.7	6.8	7.3	5.7	4.1	3.5
	5% WB	MCDBR	3.6	3.9	9.1	13.6	12.2	11.9	11.6	11.8	10.5	7.4	4.5	3.6
		MCWBR	3.5	3.4	6.5	8.1	6.5	6.5	6.2	6.4	6.8	5.6	4.0	3.5
Clear-Sky Solar Irradiance	taub	0.292	0.325	0.359	0.402	0.391	0.382	0.415	0.413	0.376	0.352	0.333	0.294	
	taud	2.189	2.166	2.201	2.211	2.313	2.367	2.294	2.293	2.370	2.400	2.269	2.212	
	Ebn at Noon	611	726	796	814	850	863	822	797	778	704	557	517	
	Edh at Noon	59	87	108	124	118	113	120	113	92	71	56	46	
All-Sky Solar Radiation	RadAvg	0.54	1.33	2.64	4.04	5.32	5.53	5.20	4.29	2.92	1.41	0.49	0.34	
	RadStd	0.12	0.26	0.43	0.49	0.51	0.50	0.47	0.40	0.31	0.26	0.08	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.48	N/A	N/A	N/A	N/A	+0.68	N/A	-175	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page